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Overview of Stark Industries Launch Systems

Simone Frescura, Paolo Mornato, Matteo Risi Mazzola, Tommaso Santoro, Lorenzo Veronese, Gabriele Pisciotta, Edouard Desmedt

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Abstract

This report presents the preliminary design of the Stark-100k, a solid rocket motor developed for nozzle throat heat-load characterization under representative operating conditions. The system is required to deliver a nominal sea-level thrust of 100 kN and a total impulse of 2.5 MN s, while employing a non-aluminized AP/HTPB composite propellant and a quasi-neutral BATES grain configuration. A distinctive feature of the design is the absence of a conventional refractory throat insert: instead, the nozzle throat region is instrumented through a water-cooled jacket, with the possible addition of a thermal barrier coating on the hot-gas side.

The design methodology combines ballistic characterization through Vieille's law regression, thermochemical analysis based on NASA CEA, and a quasi-steady chamber model to predict pressure, burning rate, thrust, and burn time. Real nozzle performance is estimated by correcting ideal thrust coefficients for divergence and viscous losses. In parallel, the thermal management system is sized through empirical heat-transfer correlations, allowing the definition of the thermal barrier coating thickness, cooling-channel geometry, coolant mass flow rate, and external pressurization architecture.

The final design satisfies the main assignment requirements and yields an initial thrust of 100 kN, a maximum chamber pressure of 70 bar, a burn time of 24.99 s, and an effective specific impulse of 240.4 s. The total impulse reaches 2.660 MN s, slightly exceeding the target because of the intrinsic mid-burn thrust hump associated with quasi-neutral BATES operation. The thermal analysis shows that the adoption of a YSZ-based thermal barrier coating is necessary to maintain acceptable wall temperatures, while the selected cooling-jacket configuration ensures safe single-phase operation with adequate thermal margins.

Overall, the proposed configuration is internally consistent and well suited for reusable ground-test applications focused on heat-transfer measurements. Although the adopted models remain simplified and conservative, they provide a robust preliminary framework and identify clear directions for future refinement toward higher-fidelity ballistic and thermal predictions.

Nomenclature

Table 1: Nomenclature of Variables and Symbols

Symbol	Definition	Unit
A	Local nozzle cross-sectional area	m^2
A_{ann}	Annular flow cross-sectional area	m^2
$A_b, A_{b,req}$	Burning surface area (effective, required)	m^2
A_e	Nozzle exit cross-sectional area	m^2
A_p	Propellant grain port area	m^2
A_t	Nozzle throat cross-sectional area	m^2
a	Vieille's law coefficient	$\mathrm{mm/s/bar^n}$
a_{SI}	Vieille's law coefficient in International System units	$\mathrm{m/s/Pa^n}$
$AR(x)$	Local nozzle area ratio (A/A_t)	-
B	Erosive burning coefficient	$\mathrm{mm/s/(m/s)^{0.8}}$
b_{ch} (or h_{ch})	Cooling channel height	mm
$C_F, C_{F,ideal}, C_{F,real}$	Thrust coefficient (generic, ideal, real)	-
C_f	Skin friction coefficient	-
C_p	Specific heat at constant pressure	$\mathrm{J/(kg \cdot K)}$
c^*	Characteristic velocity	m/s
D_{bore}	Inner grain port diameter	mm
D_e	Nozzle exit diameter	mm
D_{ext}	Grain outer diameter	mm
D_h	Cooling channel hydraulic diameter	mm
D_t	Nozzle throat diameter	mm
E, E_{wall}	Young's modulus (elasticity) of the wall material	Pa
F, F_{req}, F_{max}	Thrust (generic, required, maximum MEOP)	N
f_D	Petukhov friction factor	-
g_0	Standard gravity acceleration	$\mathrm{m/s^2}$
$h_c, h_{c,real}, h_{c,req}, h_{c,fix}$	Coolant-side convective heat transfer coefficient	$\mathrm{W/(m^2 \cdot K)}$
$h_{c,eff}$	Effective coolant-side convective heat transfer coefficient	$\mathrm{W/(m^2 \cdot K)}$
h_{fg}	Latent heat of vaporization of water	J/kg
h_g	Gas-side convective heat transfer coefficient	$\mathrm{W/(m^2 \cdot K)}$
I_{sp}	Specific impulse	s
$I_{tot}, I_{t,req}$	Total impulse (effective, required)	$\mathrm{N \cdot s}$
K_i	Concentrated pressure loss coefficients (valves, orifices)	-

Table 1 – Continued from previous page

Symbol	Definition	Unit
$k, k_{wall}, k_{TBC}, \ldots$	Thermal conductivity of various layers/fluids	$\mathrm{W/(m \cdot K)}$
L_0	Initial grain segment length	mm
L_{aft}	Aft clearance length	mm
M	Mach number	-
$M_p, M_{liner}, M_{chamber}, .$	Mass of individual components (propellant, liner, etc.)	kg
M_{mol}	Molar mass of combustion gas	kg/kmol
m	Fin parameter (for efficiency calculation)	1/m
m_{ign}	Igniter charge mass (pyrogen)	kg
m_w, m_{N_2}	Storage mass (coolant water, nitrogen)	kg
$\dot{m}, \dot{m}_{gen}, \dot{m}_{nozz}$	Mass flow rate (generic, generated, exhausted)	kg/s
$\dot{m}_w, \dot{m}_{H_2O}, \dot{m}_c$	Coolant mass flow rate in channels	kg/s
N_{ch}	Number of cooling channels	-
Nu	Nusselt number	-
n	Pressure exponent in Vieille's law	-
$P_c, P_{c,in}, P_{c,max}$	Combustion chamber pressure (current, initial, maximum)	Pa
p_0	Stagnation pressure in combustion chamber	Pa
p_c	Thermodynamic critical pressure of water	Pa
p_{dump}	Dump tank pressure	Pa
p_{inlet}	Static pressure at cooling jacket inlet	Pa
$p_{N_2,init}, p_{N_2,final}$	Nitrogen pressure in tank (initial, final lock-up)	Pa
$p_{sat}(T), p_{vap}$	Local saturation (or vapor) pressure of water	Pa
$p_{static}(x)$	Local coolant static pressure along channels	Pa
p_{tank}	Regulated water tank pressure	Pa
Pr	Prandtl number	-
Q_{tot}	Total thermal power exchanged by the jacket	W
$q, q_{flux}, q_{req}, q_{new}$	Wall heat flux (convective or total)	$\mathrm{W/m^2}$
$q_{CHF,pool}$	Critical heat flux evaluated under pool boiling conditions	$\mathrm{W/m^2}$
q_{rad}	Radiative heat flux	$\mathrm{W/m^2}$
R	Specific combustion gas constant	$\mathrm{J/(kg \cdot K)}$
R_{ext}	Propellant grain outer radius	mm
R_{N_2}	Nitrogen specific constant	$\mathrm{J/(kg \cdot K)}$
R_{tot}	Equivalent total thermal resistance	$\mathrm{m^2 \cdot K/W}$
R_c	Throat curvature radius	mm
R_e	Nozzle exit radius	mm
R_t	Nozzle throat radius	mm
Re, Re_L	Reynolds number (local in channels, longitudinal)	-
$r(x)$	Local nozzle wall radius	mm
r_i	Initial propellant grain port radius	mm
r_b	Propellant linear regression rate (burn rate)	mm/s
$\dot{r}_{insul}$	Thermal insulation linear regression rate	mm/s
$S_{cyl}, S_{fwd}, S_{aft}$	Chamber internal surfaces (cylindrical, domes)	$\mathrm{m^2}$
$SF_{cav}, SF_{CHF}, SF_{mec}$	Safety factors (cavitation, CHF, mechanical)	-

Table 1 – Continued from previous page

Symbol	Definition	Unit
T_0	Stagnation temperature in combustion chamber	K
T_c	Thermodynamic critical temperature of water	K
T_s	Local static gas temperature	K
$T_{aw}, T_{aw,eff}$	Adiabatic wall temperature (standard and effective)	K
T_{bond}	Metallic bond coat interface temperature	K
T_{gas}	Initial pressurant gas temperature (nitrogen)	K
$T_{hg}, T_{surface}$	Gas-side surface temperature	K
$T_{lim,wall}, T_{lim,TBC}$	Operating thermal limit (copper wall, TBC)	K
$T_{sat}(p)$	Local water saturation temperature	K
$T_w(x), T_{w,bulk}, T_{in}$	Coolant bulk temperature (local, inlet)	K
T_{wc}	Coolant-side wall temperature	K
T_{wg}	Gas-side copper wall temperature (under TBC)	K
t	Generic time	s
t_b	Motor burn time	s
t_{op}	Total cooling system operating time	s
t_{bond}	Bond coat layer thickness	m
t_{fin}	Cooling channel fin thickness	m
t_{insul}, t_{liner}	Internal insulation and elastomeric liner thickness	m
t_{TBC}	Thermal Barrier Coating (YSZ) thickness	m
t_{wall}	Structural wall thickness (CuCrZr liner)	m
V	Gas velocity in grain port	m/s
v_{water}, v_w	Coolant velocity in channels	m/s
V_F	Motor internal free volume	m^3
V_w, V_{N_2}	Useful tank volumes (water, nitrogen)	m^3
w	Propellant grain web thickness	mm
w_{AP}, w_{HTPB}	Weight fractions of Oxidizer (AP) and Binder (HTPB)	-
$w_{ch}(x)$	Local cooling channel width	m
x	Axial coordinate along the nozzle	m
α	Nozzle divergent section half-angle	rad
β	Nozzle convergent section half-angle	rad
γ_c, γ_{N_2}	Ratio of specific heats (combustion gas, nitrogen)	-
ξ	Linear thermal expansion coefficient	1/K
$\Delta C_{F,visc}$	Viscous friction loss contribution	-
Δp	Differential pressure between coolant and hot gas	Pa
$\Delta p_{feedline}$	Total pressure drop along the feed line	Pa
$\Delta p_{conc}, \Delta p_{dist}$	Localized pressure drops (concentrated, distributed)	Pa
$\Delta T_{ONB,req}$	Wall superheat required to initiate boiling	K
ΔT_{sup}	Effective wall superheat ($T_{wc} - T_{sat}$)	K
ΔT_{wall}	Radial thermal gradient across the structural wall	K

Table 1 – Continued from previous page

Symbol	Definition	Unit
ϵ, ϵ_c	Expansion ratio (exit/throat) and contraction ratio (chamber/throat)	-
ϵ_g	Effective gas emissivity	-
η_{fin}	Channel fin thermal efficiency	-
η_o	Global efficiency of the finned surface	-
θ	Dimensionless temperature parameter in Wagner-Pruss eq.	-
λ	Thrust correction factor for divergence loss	-
μ, μ_0, μ_e	Dynamic viscosity (local, stagnation, exit)	$\mathrm{Pa} \cdot \mathrm{s}$
ν	Structural material Poisson's ratio	-
ρ, ρ_{water}	Local coolant density	$\mathrm{kg/m^3}$
ρ_c	Stagnation density in combustion chamber	$\mathrm{kg/m^3}$
$\rho_{insul}, \rho_{liner}$	Thermal insulation material density	$\mathrm{kg/m^3}$
$\rho_p, \rho_{AP}, \rho_{HTPB}$	Propellant density (bulk, AP crystals, HTPB binder)	$\mathrm{kg/m^3}$
ρ_V	Water vapor density at local pressure	$\mathrm{kg/m^3}$
σ	Bartz boundary layer correction factor	-
σ_p	Maximum primary mechanical stress in the wall	Pa
$\sigma_y(T)$	Temperature-degraded yield strength	Pa
σ_{bend}	Bending stress at fin root	Pa
σ_{LG}	Liquid-gas surface tension (water)	N/m
σ_{mem}	Membrane stress (hoop stress)	Pa
σ_{SB}	Stefan-Boltzmann constant	$\mathrm{W/(m^2 \cdot K^4)}$
σ_{th}	Biaxial thermal stress	Pa
σ_{VM}	Von Mises equivalent stress	Pa

List of Acronyms

BATES : Ballistic Test and Evaluation System

YSZ : Yttria Stabilized Zirconia

TBC : Thermal Barrier Coating

APS : Atmospheric Plasma Spray

RSZ : Rare-earth Stabilized Zirconia

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Chapter 1

Introduction

The present report documents the preliminary design of the Stark-100k, a solid propellant rocket motor developed by Stark Industries for nozzle throat heat load characterization. The motor is designed to deliver 100 kN of thrust at sea level with a total impulse of 2.5 MN·s, using a non-aluminized AP/HTPB composite propellant.
A distinctive feature of this motor is that the nozzle throat region does not employ a conventional ablative or refractory insert. Instead, a water-cooled jacket covers the nozzle from area ratio 2 (subsonic side) through the throat to area ratio 2 (supersonic side), enabling steady-state heat transfer measurements. A thermal barrier coating may be applied on the hot-gas side of the jacket to reduce the thermal load on the metallic structure; its necessity is assessed quantitatively in this work.

1.1 Requirements Summary

Table 1.1 summarizes the top-level requirements driving the design.

Table 1.1: Top-level design requirements.

Parameter	**Requirement**	**Value**
Thrust (sea level)	Nominal	100 kN
Total impulse	Target	2.5 MN·s
Chamber pressure	MEOP	70 bar
Propellant	Non-aluminized	AP/HTPB
Thrust profile	Quasi-neutral	BATES grain
Nozzle expansion	Optimal at SL	$P_e = P_a$
Cooling jacket coverage	Fixed	$\varepsilon = 2$ to $\varepsilon = 2$
Coolant	Pressurized water	$T_{in} = 18\,^{\circ}\mathrm{C}$

Since the primary objective is heat load quantification rather than flight performance, the motor is not necessarily optimized for mass or volumetric efficiency. Design choices such as a single-segment BATES grain and a simple conical nozzle prioritize manufacturing simplicity and predictability of the internal ballistic environment over performance.

Chapter 2

Literature Review

2.1 Thermal Barrier Coatings (TBCs) and Plasma Spray Processing

Thermal Barrier Coatings (TBCs) are essential components in modern aerospace and industrial gas turbine applications, designed to increase engine efficiency and component durability by allowing higher operating temperatures while maintaining lower substrate temperatures [12]. In the specific context of rocket thrust chambers, TBCs must withstand extreme thermal transients and high heat fluxes, often requiring specialized materials and deposition techniques.

Thermal Barrier Coatings (TBCs) typically consist of a multi-layer system: a ceramic top layer—such as the advanced Rare-earth-Stabilized Zirconia (RSZ) or the standard Yttria-Stabilized Zirconia (YSZ)—and a metallic bond coat. While the top ceramic layer provides the primary thermal insulation against high heat fluxes, the metallic bond coat (often NiCrAlY) serves a dual purpose: it ensures the mechanical adhesion of the ceramic to the substrate wall and protects the underlying metal from oxidation. Furthermore, the bond coat acts as a critical strain-buffer, accommodating the mismatch in the Coefficient of Thermal Expansion (CTE) between the structural substrate and the ceramic overcoat, which is essential for maintaining coating integrity during the rapid thermal transients characteristic of rocket thrust chambers.

While conventional Yttria-Stabilized Zirconia (YSZ) is the industry standard, it often fails quickly above 1600 °C in rocket engine applications [10]. To address this, double-ceramic-layer (DCL) architectures have been developed, such as the RSZ/YSZ system. This quaternary RSZ topcoat demonstrates superior thermal shock resistance, achieving up to 15 cycles at 1800 °C [10]. Failure is typically governed by sintering-induced localized spallation rather than immediate substrate deformation. Spallation is caused by the differing thermal expansion coefficients of the various layers, which causes the top layer and the bond coat to expand at different rates and eventually detach from one another. Additionally, failure is often associated with the intermediate bond layer between the substrate and the protective top ceramic coat. Experimental evidence indicates that under constant stress, this layer is prone to oxidation as air penetrates the inherent porosity of the overlying ceramic layer. This progressive oxidation ultimately leads to the separation of the top layer from the bond coat. Analytical techniques, including finite element and finite-difference models, are used to predict the thermomechanical behavior of TBCs. In high heat flux environments, extremely thin coatings (sometimes < 0.02

mm) are required to maintain structural integrity [34]. Modeling indicates that a 100 μm thick $ZrO_2 - Y_2O_3$ coating can provide a metal temperature benefit of approximately 300 °C, while a 200 μm coating can provide a benefit exceeding 500 °C [12]. The geometry of the substrate also plays a vital role; for instance, coatings on rod substrates tend to be more durable than those on tubes because the greater thermal mass of the rod results in lower temperatures during both deposition and testing [12]. The literature confirms that while material choice (like RSZ/YSZ) and thickness are critical for performance [10], [12], the deposition process, particularly the use of N_2 plasma gas, offers a robust path toward more efficient and cost-effective manufacturing [9].
The atmospheric plasma spray (APS) process is widely used to deposit zirconia-based coatings. Traditional methods typically utilize argon-hydrogen ($Ar - H_2$) plasma gas mixtures, but research into nitrogen-hydrogen ($N_2 - H_2$) plasmas has revealed significant process advantages. Specifically, using N_2 as the primary plasma gas can result in higher in-flight particle temperatures compared to Ar, although particle velocities tend to be lower [9]. A critical finding is that deposition efficiency (DE) can be doubled when using N_2 instead of Ar, reaching values in the range of 60-70% [9]. This suggests a compelling economic case for nitrogen-based spraying due to reduced powder consumption and higher production rates. Furthermore, as-sprayed coatings from both gas mixtures exhibit similar levels of porosity and hardness [9].

2.2 BATES Motor Principles

The BATES (Ballistic Test and Evaluation System) grain configuration main feature is a cylindrical segment with an internal circular bore, where both the radial bore surface and the two axial end faces burn simultaneously. This combination of progressive (bore) and regressive (end face) contributions produces a quasi-neutral thrust profile where initial and final pressures are equal, with a moderate hump at mid-burn. The BATES geometry has been widely adopted in small- and medium-scale test motors precisely because of this quasi-neutral behavior, which simplifies the extraction of ballistic parameters from static firing data [3, 4]. The 15 data points provided in the assignment were obtained following this approach: each point corresponds to a small-scale BATES motor fired at a controlled pressure level, from which the effective burning rate was derived via a standard regression algorithm. Beyond propellant characterization, the BATES configuration has found application in flight-weight motors where a nearly constant thrust is required without the complexity of star or finocyl grain geometries [4]. Its main advantage is geometric simplicity: the neutrality condition depends on a single constraint relating segment length, bore radius, and web thickness ($L_0 = 4r_i + 3w$) [7], making analytical sizing straightforward. The primary limitation is that exact neutrality holds only at the endpoints of the burn; the mid-burn pressure peak (typically 5–15% above the nominal value, depending on the pressure exponent n) must be accounted for in the structural sizing of the motor case. The quasi-steady internal ballistics model used in this work where the chamber pressure is determined at each instant by the equilibrium between mass generation and nozzle discharge follows the standard formulation described in [3] and in [7]. The detailed equations and their implementation are presented in Section 3.2.

Chapter 3

Methodology

3.1 Solid Propellant System Design

3.1.1 Propellant Composition

The Stark-100k motor employs a non-aluminized AP/HTPB composite propellant, with ammonium perchlorate as oxidizer and hydroxyl-terminated polybutadiene as binder in a mass ratio of 84.18/15.82. The corresponding oxidizer-to-fuel ratio is $O/F = 5.32$. This composition was selected because it satisfies the design requirements with an adequate margin when combined with the other chosen design parameters. The propellant bulk density was then evaluated from the component densities using the ideal mixing rule:

$$\rho_p = \frac{1}{\dfrac{w_{AP}}{\rho_{AP}} + \dfrac{w_{HTPB}}{\rho_{HTPB}}} \approx 1654 \text{ kg/m}^3 \tag{3.1}$$

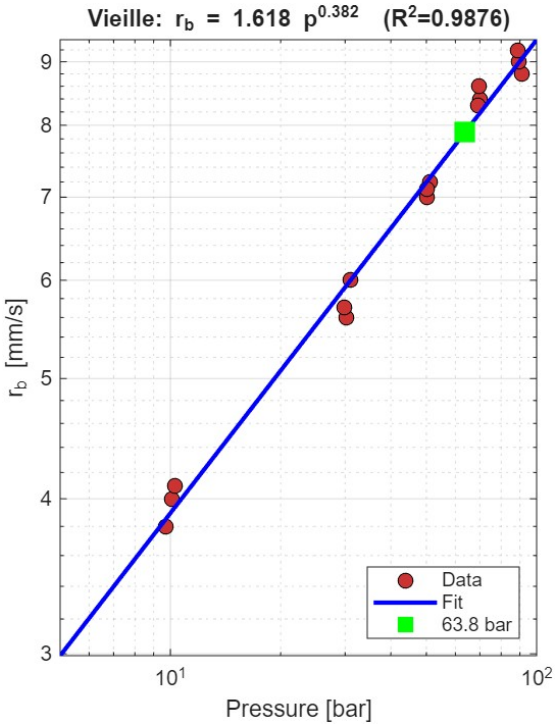
where $\rho_{AP} = 1950\,\text{kg/m}^3$ and $\rho_{HTPB} = 915\,\text{kg/m}^3$.

3.1.2 Ballistic Properties: Vieille's Law

The burning rate law was derived from the 15 experimental data points provided in the assignment, covering the pressure range 10–91 bar. A power-law fit of Vieille's form was performed in log-log space:

$$r_b = a\,P_c^{\,n} \tag{3.2}$$

where r_b is in mm/s and P_c in bar. The linear regression $\ln(r_b) = n\ln(P_c) + \ln(a)$ yielded: $a = 1.6177$ mm/s/bar^n, $n = 0.3816$, $R^2 = 0.9876$.
The coefficient of determination $R^2 \approx 0.988$ confirms the validity of the power-law assumption over the tested pressure range.



For use in the quasi-steady chamber model, the Vieille coefficient was converted to SI units (r_b in m/s, P_c in Pa):

$$a_{SI} = a \times 10^{-3} \times (10^5)^{-n} \approx 2.00 \times 10^{-5} \text{ m/s/Pa}^n \tag{3.3}$$

3.1.3 Thermochemistry: NASA CEA

The thermochemical properties of the combustion products were obtained from a NASA CEA run using the following inputs:

- Fuel: C_4H_6 (1,3-butadiene; which, for a preliminary analysis, can be used as a HTPB proxy, as shown in [1])
- Oxidizer: NH_4ClO_4(I) (ammonium perchlorate, solid phase)
- $O/F = 5.32$
- $P_c = 70\,\text{bar}$
- $P_c/P_e = 69.1$ (optimal expansion at sea level)

The equilibrium model was applied from the combustion chamber to the throat, while a frozen composition assumption was used from the throat through the nozzle expansion. This approach accounts for chemical equilibrium up to the subsonic condition, where reaction rates are still sufficiently fast to maintain equilibrium, and then assumes frozen chemistry in the divergent section, where flow expansion occurs too rapidly for significant recombination to take place. This hybrid model was selected instead of a fully equilibrium or fully frozen approach because it provides a more realistic representation of the actual nozzle physics, balancing the over-optimistic performance prediction of a fully equilibrium model with the overly conservative nature of a fully frozen assumption. The key results from CEA are summarized in Table 3.1.

Table 3.1: NASA CEA results (chamber conditions)

Parameter	**Value**	**Unit**
Combustion temperature, T_c	2852.3	K
Molar mass, M_{mol}	23.683	kg/kmol
Ratio of specific heats, γ_c	1.1953	–
Gas density in chamber, ρ_c	6.6911	kg/m^3
Specific gas constant, R	351.07	J/(kg·K)
Characteristic velocity, c^*	1542.0	m/s
Ideal thrust coefficient, $C_{F,ideal}$	1.5754	–
Optimal expansion ratio, ε	8.1223	–
Exit viscosity, μ_e	5.315×10^{-5}	Pa·s

3.1.4 Quasi-Steady Chamber Model

The chamber pressure during the burn is governed by the instantaneous equilibrium between the mass generated by propellant combustion and the mass discharged through the

nozzle throat. Setting $\dot{m}_{gen} = \dot{m}_{nozz}$, i.e., $\rho_p\, a_{SI}\, P_c^n\, A_b = P_c\, A_t/c^*$, and solving for P_c yields:

$$P_c(x) = \left(\frac{\rho_p\, a_{SI}\, A_b(x)\, c^*}{A_t} \right)^{1/(1-n)} \tag{3.4}$$

where all quantities are in SI units and P_c is obtained in Pa. This expression provides the chamber pressure for each value of the burning surface, where x represents the burned thickness of the grain. To obtain the evolution of the chamber pressure, we perform an integration over time with a fixed step of $\Delta t = 0.01\,\text{s}$ while exploiting the following equations:

$$r_b(x) = a_{SI}\, P_c(x)^n \tag{3.5}$$

$$x_{i+1} = x_i + r_b\, \Delta t \tag{3.6}$$

Time integration is carried out iteratively. Starting from the initial burning surface and the initial pressure, we compute the initial burning rate. The burning surface at the next time step is updated through Eq. (3.6). The corresponding chamber pressure and burning rate are then computed using Eq. (3.4) and Eq. (3.5), respectively. Repeating this procedure provides the evolution of chamber pressure over time. A parametric analysis was also performed in NASA CEA by varying chamber pressure between 60 and 80 bar, a range deliberately wider than the expected operating conditions. The results showed that the characteristic velocity c^* changes only by a few percent, indicating a weak dependence on chamber pressure compared with mixture ratio. For this reason, c^* was assumed constant in the quasi-steady chamber model. Within the scope of a preliminary design study, this approximation is considered adequate, since a pressure-dependent formulation would increase model complexity without a significant gain in accuracy.

3.1.5 Ideal vs Real Performance Parameters

The thrust coefficient from NASA CEA assumes ideal isentropic expansion and neglects losses such as flow divergence and viscous effects; these were included in the model via correction factors to obtain a more realistic estimate. The nozzle geometry was defined with a 45° half-angle in the convergent section to ensure a compact configuration. For the divergent section, a 15° half-angle was selected, as this is a widely adopted engineering compromise between nozzle performance and geometrical constraints, consistent with the range commonly recommended in the literature [3].
The real thrust coefficient, accounting for the divergent half angle stated before and integrated skin friction (which will be better explained in section 3.2.1) along the nozzle wall, is:

$$C_{F,real} = C_{F,ideal} \cdot \lambda - \Delta C_{F,visc} = 1.5288 \tag{3.7}$$

where $\lambda = (1 + \cos\alpha)/2 = 0.9830$ is the divergence loss factor and $\Delta C_{F,visc} = 0.0197$ is the viscous drag contribution obtained by integrating the local skin friction over the discretized nozzle wall using CEA-derived density and velocity profiles. The resulting effective specific impulse is:

$$I_{sp} = \frac{C_{F,real} \cdot c^*}{g_0} = 240.4\ \text{s} \tag{3.8}$$

In the present analysis, the thrust coefficient $C_{F,ideal}$ was assumed to be constant. Similarly to the approach adopted for c^*, this assumption was justified by means of NASA CEA calculations, which showed that the variation of $C_{F,ideal}$ over the 60 to 80 bar chamber-pressure range is negligible.

3.2 Grain and Nozzle Geometry

3.2.1 Nozzle Design

As stated in Section 3.1.5, a conical nozzle configuration was adopted with a divergent half-angle $\alpha = 15°$ and a convergent half-angle $\beta = 45°$. The optimal expansion ratio $\varepsilon = A_e/A_t = 8.1223$ was obtained from NASA CEA under an equilibrium formulation up to the throat, with frozen flow assumptions applied from the throat to the nozzle exit, corresponding to optimal expansion at sea level conditions ($P_e = P_a$).

- **Divergence loss**: the conical nozzle produces a non-axial exhaust, reducing the thrust by a factor $\lambda = (1 + \cos\alpha)/2 = 0.9830$.
- **Viscous friction**: the skin friction drag was computed by discretizing the nozzle wall into 150 elements (both convergent and divergent) and integrating the local shear stress using CEA-derived density and velocity profiles at each area ratio. The flat-plate skin friction coefficient $C_f = 0.074/Re_L^{0.2}$ was applied with the total nozzle length as the reference, yielding $Re_L = 5.22 \times 10^6$.
- **Two-phase flow losses**: two-phase flow losses were neglected and set to zero. As discussed in the literature [2], these losses are mainly associated with the kinematic and thermal lag of large condensed particles, such as aluminum oxide agglomerates. Since the Stark-100k uses a non-aluminized AP/HTPB propellant, such particles are absent in the exhaust, making this loss mechanism negligible.

3.2.2 BATES Grain Design Philosophy

As reported in 1.1, single-segment BATES (Ballistic Test and Evaluation System) grain configuration was selected to prioritize manufacturing simplicity. In a BATES grain, three surfaces burn simultaneously: the cylindrical bore surface (radially outward) and the two flat end faces (axially inward). Under the assumption that the instantaneous burning rate is equal for all of these surfaces, then for any burned thickness x the total burning area can be written as:

$$A_b(x) = \underbrace{2\pi(r_i + x)(L_0 - 2x)}_{\text{cylindrical bore}} + \underbrace{2\pi\left[R^2 - (r_i + x)^2\right]}_{\text{two end faces}} \tag{3.9}$$

where r_i is the initial bore radius, R is the outer grain radius, and L_0 is the initial segment length.
The BATES neutrality condition requires the burning area at the beginning and at the end of combustion to be equal, i.e., $A_b(0) = A_b(w)$, where $w = R - r_i$ is the web thickness. This condition, derived analytically from Eq. (3.9), yields:

$$L_0 = 4\,r_i + 3\,w \tag{3.10}$$

Under this constraint, and substuting Eq.(3.9) in Eq.(3.4), it can be seen that the pressure profile, as a function of x, exhibits a symmetric hump: P_c starts at $P_{c,in}$, rises to a maximum $P_{c,max}$ at mid-burn ($x = w/2$), and returns to approximately $P_{c,in}$ at burnout. The resulting thrust profile is quasi-neutral, with a peak-to-nominal ratio governed by the pressure exponent n.

The residual difference between initial and final chamber pressure ($\Delta P_c = 0.009$ bar) is attributable to the finite time step $\Delta t = 0.01s$, which prevents the last computed point from falling exactly at $x = w$. This discrepancy vanishes as $\Delta t \to 0$ and does not affect the design results.

3.2.3 Iterative Design Procedure

The grain geometry was determined through a three-level nested bisection algorithm, which simultaneously satisfies three constraints:

1. **Outer bisection on** $P_{c,in}$: finds the initial chamber pressure such that the maximum pressure during the burn equals the MEOP target ($P_{c,max} = 70.0\,\text{bar}$).

2. **Nozzle sizing from** F_{req} **and** $I_{t,req}$: the required thrust $F_{req} = 100\,\text{kN}$ sets the throat area A_t (iterated until $F = P_c\,A_t\,C_{F,real} = F_{req}$), while the required total impulse $I_{t,req} = 2.5\,\text{MN·s}$ defines the burn time $t_b = I_{t,req}/F_{req} = 25\,\text{s}$, which drives the web thickness sizing.

3. **Middle bisection on web** w: for a given $P_{c,in}$, finds the web thickness such that the integrated burn time equals the required value $t_b = I_t/F = 25\,\text{s}$, using the quasi-steady time-marching scheme of Eqs. (3.4)–(3.6).

4. **Inner bisection on** r_i/R: for a given web, finds the bore-to-outer radius ratio such that $A_b(0) = A_{b,req}$, where $A_{b,req} = \dfrac{A_t}{\rho_p\,a_{SI}\,c^*}\,P_{c,in}^{1-n}$ is the burning area required to produce the initial calculated chamber pressure $P_{c,in}$.

3.2.4 Chamber Clearances

Two clearance volumes separate the grain from the forward and aft closures:

Forward clearance (400 mm). This space accommodates the pyrogen igniter assembly, selected from the Antares III (in literature [13] [14]) due to its comparable requirements, and having an axial length of 356 mm. The igniter charge mass was estimated using the Sutton correlation [3]:

$$m_{ign} = 0.12\,V_F^{0.7} \approx 112\ \text{g} \tag{3.11}$$

where $V_F = \pi r_i^2 L_0 \approx 17\,483\,\text{in}^3$ is the motor free volume (in cubic inches, as per the original formulation). An additional 50 mm margin is included for structural mounting hardware.

Aft clearance (180 mm). This distance ensures a smooth gas flow transition from the cylindrical bore to the convergent nozzle entrance. From mass continuity, the flow velocity is inversely proportional to the available cross-sectional area. If the annular passage downstream of the grain is too restrictive, the flow undergoes a rapid acceleration, leading to increased shear stress, pressure losses, and potential erosion at the grain aft end. To mitigate these effects, the annular flow area

$$A_{ann} = 2\pi r_i L_{aft}$$

is required to exceed the port area

$$A_p = \pi r_i^2$$

by a safety factor of 1.5, chosen from a sensitivity analysis which shows that values below 1.3 lead to noticeable velocity amplification, supporting the adoption of a conservative margin. The selected value represents a trade-off between robustness and geometric efficiency. The resulting sizing condition is:

$$L_{aft} \geq 1.5 \times \frac{r_i}{2} = 1.5 \times \frac{241}{2} \approx 180 \text{ mm} \tag{3.12}$$

This prevents flow acceleration and localized erosion at the grain-to-nozzle transition region.

3.2.5 Thermal Protections

Combustion Chamber Insulation

The combustion chamber case is protected by EPDM (ethylene propylene diene monomer) insulation filled with silica ($\rho_{insul} = 1200\,\text{kg/m}^3$, $k = 0.30\,\text{W/(m·K)}$, $T_{dec} = 700\,\text{K}$) [16, 17], applied at uniform thickness per zone. The insulation thickness is sized for the nominal exposure time (full burn duration $t_b \approx 25\,\text{s}$) using regression rates from the literature [3, 4]:

$$t_{insul} = \dot{r}_{insul} \times t_b \times SF \tag{3.13}$$

where $SF = 1.5$ is a safety factor chosen according to the same rationale discussed in 3.2.4. The regression rates vary by zone depending on the local flow conditions: $0.08mm/s$ for the forward dome (low gas velocity), $0.10mm/s$ for the cylindrical wall, and $0.15mm/s$ for the aft dome where the gas converges toward the nozzle producing higher turbulence.

Liner

A thin HTPB-based liner (1.0 mm thickness) is applied between the propellant grain and the EPDM insulation. Its functions are twofold: ensuring structural adhesion (case bonding) and acting as a migration barrier for propellant plasticizer. The liner thickness is consistent with production data for the Ariane 5 solid rocket boosters [5]. The liner composition, HTPB with a DDI curing agent, DOS plasticizer, and carbon black filler, is chemically compatible with both the HTPB-based propellant and the EPDM insulation [6]. The estimated liner mass is:

$$M_{liner} = \rho_{liner} \cdot (S_{cyl} + S_{fwd} + S_{aft}) \cdot t_{liner} \approx 950 \times 5.47 \times 0.001 \approx 5.2 \text{ kg} \tag{3.14}$$

Nozzle Ablative Protection

The nozzle sections not covered by the ceramic insert which extends from $\varepsilon = 2$ through the throat on both sides are protected by the same EPDM + silica material used in the combustion chamber. The procedure is to get the wall length not covered by ceramic insulation by basic trigonometric calculations, then use the (3.13).

3.2.6 Erosive Burning Assessment

Erosive burning was assessed using the formulation shown in [7]:

$$r_b = a\,P_c^n + B\,V^{0.8} \tag{3.15}$$

where V is the gas velocity in the port and $B \approx 5 \times 10^{-4}$ mm/s/(m/s)$^{0.8}$ is the erosive burning coefficient for AP/HTPB propellants. The port-to-throat area ratio at ignition is:

$$\frac{A_p}{A_t} = \frac{\pi r_i^2}{A_t} \approx 17.8 \tag{3.16}$$

This is well above the threshold of 4 cited in the literature [3], below which erosive burning becomes significant. The estimated port velocity is $V \approx 37\,\text{m/s}$, producing an erosive contribution of approximately 0.01 mm/s, less than 0.12% of the total burning rate. Erosive burning is therefore negligible and was not included in the final performance model.

3.3 Cooling Jacket and Thermal Protection

3.3.1 Thermal Analysis Methodology

A comprehensive sizing algorithm was employed to dimension the cooling channels(described in 3.4.5 – 3.4.8). Subsequently, a separate high-fidelity 1D thermo-fluid marching solver was developed to rigorously verify the heat loads and ensure the test article's survivability. The verification model integrates the longitudinal equations node-by-node ($N = 200$ axial stations), coupling the hot-gas aerodynamics, the multi-layer solid conduction, and the coolant hydraulics.

3.3.2 Geometry and Material Architecture

The basic nozzle profile features a conical convergent section (45°) and a conical divergent section (15°). To prevent non-physical geometric cusps at the throat, which would trigger analytical singularities and artificial spikes in the heat transfer coefficient calculation, a circular arc rounding with a curvature ratio of $R_c = 1.5R_t$ was implemented, ensuring geometric continuity. The cooling jacket follows a milled-channel configuration [22]. To accommodate the varying nozzle circumference while maintaining a constant fin thickness ($t_{fin} = 1.5$ mm) for uniform structural integrity, the channel width $w_{ch}(x)$ varies continuously along the axial coordinate x:

$$w_{ch}(x) = \frac{2\pi r(x)}{N_{ch}} - t_{fin} \tag{3.17}$$

The primary structural liner is composed of a Copper-Chromium-Zirconium alloy (CuCrZr). Theoretically, a high thermal conductivity substrate ($k \approx 320$ W/(mK)) is essential to rapidly conduct heat away to the coolant, flattening the thermal gradient across the metal wall and avoiding localized hot spots. CuCrZr is specifically chosen over standard Oxygen-Free High Thermal Conductivity (OFHC) copper due to its precipitation-hardening properties, which grant it superior yield strength retention at elevated temperatures [23]. However, to keep the copper substrate strictly below its over-ageing metallurgical limit (773 K) while operating at extreme heat fluxes, a Thermal Barrier Coating

(TBC) is mandatory. Yttria-Stabilized Zirconia (YSZ) is applied as the top coat because its exceptionally low thermal conductivity ($k_{TBC} \approx 1.5$ W/(mK)) creates a massive thermal resistance, forcing a severe temperature drop across a microscopic layer [24]. Finally, to ensure proper mechanical adhesion and mitigate the Coefficient of Thermal Expansion (CTE) mismatch between the ceramic YSZ and the metallic CuCrZr, a 35 μm NiCoCrAlY metallic bond coat is layered between them.

3.3.3 Fluid Properties Modeling: Axial Marching Assumptions

A core feature of the solver is that fluid properties are not assumed constant, but are updated at every axial step dx to capture the extreme thermodynamic gradients typical of rocket engines.

Combustion Gas: Thermodynamic and transport properties (γ, C_p, Pr, μ) for the non-aluminized AP/HTPB propellant ($O/F = 5.32$, $p_0 = 70$ bar) were computed via NASA CEA [25]. A hybrid kinetic approach was adopted depending on the axial position:

- **Convergent Section ($x \leq 0$):** Equilibrium flow is assumed. Gas velocities in the chamber and convergent section are relatively low, allowing sufficient residence time for chemical recombination as the temperature drops.
- **Divergent Section ($x > 0$):** Frozen flow is assumed. The rapid supersonic expansion kinetically "freezes" the chemical composition at throat conditions, as the fluid residence time becomes significantly shorter than the chemical reaction timescale.

Properties are linearly interpolated against the local nozzle Area Ratio ($AR(x)$).

Coolant (Water) and Saturation Equations: To accurately capture the steep fluid property variations near the wall, density (ρ) and specific heat (C_p) are calculated using the IAPWS-IF97 Region 1 standard formulation [26], which is based on a 34-term dimensionless Gibbs free energy equation $\gamma(\pi, \tau)$. These properties are evaluated at the local bulk temperature $T_w(x)$ and the local static pressure $p_{static}(x)$. Since the static pressure drops axially due to friction, the thermodynamic state of the water is continuously updated to monitor exact saturation boundaries. The vapor pressure $p_{sat}(T)$ is computed using the highly accurate Wagner-Pruss (1993) analytical relation [27]:

$$\ln\left(\frac{p_{sat}}{p_c}\right) = \frac{T_c}{T}\left(a_1\theta + a_2\theta^{1.5} + a_3\theta^3 + a_4\theta^{3.5} + a_5\theta^4 + a_6\theta^{7.5}\right) \tag{3.18}$$

where $\theta = 1-(T/T_c)$, the critical temperature is $T_c = 647.096$ K, and the critical pressure is $p_c = 22.064$ MPa. The specific coefficients used are $a_1 = -7.8595$, $a_2 = 1.8440$, $a_3 = -11.7866$, $a_4 = 22.6807$, $a_5 = -15.9618$, and $a_6 = 1.8012$. Because the Wagner-Pruss equation is explicit for pressure but implicit for temperature, the local saturation temperature $T_{sat}(p)$ is found by numerically inverting this equation via a Newton-Raphson root-finding algorithm. At each pressure node, the solver defines a residual function $f(T) = p_{sat}(T) - p_{static} = 0$ and iteratively updates the temperature guess using its derivative, $T_{n+1} = T_n - f(T_n)/f'(T_n)$, until the residual converges to zero.

3.3.4 The 1D Marching Solver and Heat Transfer Model

Because the hot-gas boundary layer properties depend on the wall temperature, and the coolant viscosity depends on the cold-wall temperature, the thermal resistances cannot be computed a priori. An explicit forward-Euler marching algorithm is implemented, featuring a Fixed-Point iteration loop at each node i.

Step 1: Coolant Hydraulic Coupling

At each node i, the local static pressure of the coolant ($p_{static,i}$) is calculated by subtracting the distributed friction losses of the previous element using the local Petukhov friction factor (f_D) [28]. This ensures precise calculation of local phase-change margins and accurate IAPWS property extraction.

Step 2: Gas-Side Convection (The Bartz Formulation)

The gas-side convective heat transfer coefficient (h_g) is evaluated using the semi-empirical Bartz equation [29]:

$$h_g = \left[\frac{0.026}{D_t^{0.2}}\left(\frac{\mu_0^{0.2}C_p}{Pr^{0.6}}\right)\left(\frac{p_0}{c^*}\right)^{0.8}\left(\frac{D_t}{R_c}\right)^{0.1}\right]\left(\frac{A_t}{A}\right)^{0.9}\sigma \tag{3.19}$$

The use of this specific formulation is built upon several fundamental approximations: (1) the core flow is treated as a 1D isentropic expansion, (2) the heat transfer mechanism is analogous to a fully developed turbulent flow in a pipe, and (3) the thermal boundary layer is assumed to begin its development at the stagnation chamber. These assumptions intrinsically overestimate the convective heat transfer, particularly in the divergent section, aligning perfectly with our conservative approach. To account for the severe property variations across the thermal boundary layer (due to the steep gradient between the combustion core and the relatively cold wall), the correlation is corrected by the factor σ. This factor must be evaluated iteratively at the unknown hot-gas-facing surface temperature (T_{hg}):

$$\sigma = \left[\frac{T_{hg}}{2T_0}\left(1+\frac{\gamma-1}{2}M^2\right)+0.5\right]^{-0.68}\left[1+\frac{\gamma-1}{2}M^2\right]^{-0.12} \tag{3.20}$$

Step 3: Gas-Side Radiative Heat Flux

In addition to forced convection, a radiative heat flux (q_{rad}) component is integrated. Unlike heavily metalized propellants (e.g., aluminized solids) which radiate intensely as optically thick clouds of condensed particles, the non-aluminized AP/HTPB solid propellant generates a relatively clean exhaust [3]. Its thermal radiation is predominantly governed by the specific emission bands of gaseous participating media, specifically CO_2 and H_2O.

To model this without resorting to complex line-by-line spectral integration, a conservative effective gray-gas emissivity of $\epsilon_g \approx 0.10$ is assigned [3]. Using the Stefan-Boltzmann law evaluated at the local static gas temperature (T_s), the radiative flux is defined as:

$$q_{rad} = \epsilon_g \sigma_{SB}(T_s^4 - T_{hg}^4) \tag{3.21}$$

To seamlessly integrate this non-linear flux into the linear 1D thermal resistance network, it is absorbed into an effective driving adiabatic wall temperature:

$$T_{aw,eff} = T_{aw} + \frac{q_{rad}}{h_g} \tag{3.22}$$

Step 4: Coolant-Side Convection (Gnielinski vs Dittus-Boelter)

While the Dittus-Boelter correlation ($Nu = 0.023Re^{0.8}Pr^{0.4}$) [30] has been intentionally employed for the geometric sizing, due to its easier integrability, it lacks the fidelity required for this detailed verification phase. It underestimates boundary layer friction variations and is inaccurate for fluids with high Prandtl numbers (such as highly subcooled water, $Pr \approx 3-7$) or in transition flow regimes ($Re \approx 3000-10000$). To rigorously verify thermal limits without risking an overestimation of the cooling jacket's capabilities, the high-fidelity **Gnielinski correlation** [31] is implemented in this solver:

$$Nu = \frac{(f_D/8)(Re - 1000)Pr}{1 + 12.7\sqrt{f_D/8}(Pr^{2/3} - 1)} \tag{3.23}$$

This formulation is vastly superior for validation because it explicitly ties the heat transfer to the hydrodynamics by incorporating the local Petukhov friction factor (f_D). The base convective coefficient ($h_c = Nu \cdot k_{water}/D_h$) is subsequently penalized by 5% where the channel aspect ratio exceeds 3, adhering to the Shah & London empirical corrections for deep rectangular channels [32]. The coefficient is then enhanced by the lateral fins. Assuming an adiabatic fin tip, the fin efficiency (η_{fin}) and overall surface efficiency (η_o) yield an effective cold-side coefficient mapped to the hot-gas surface area:

$$m = \sqrt{\frac{2h_c}{k_{wall}t_{fin}}}, \quad \eta_{fin} = \frac{\tanh(mh_{ch})}{mh_{ch}}, \quad h_{c,eff} = h_c \cdot \eta_o \cdot \frac{N_{ch}P_{wet}}{2\pi r_{local}} \tag{3.24}$$

Step 5: Conduction and Energy March

The total thermal resistance (R_{tot}) places the convective boundaries and the solid multi-layer wall in series:

$$R_{tot} = \frac{1}{h_g} + \frac{t_{TBC}}{k_{TBC}} + \frac{t_{bond}}{k_{bond}} + \frac{t_{wall}}{k_{wall}} + \frac{1}{h_{c,eff}} \tag{3.25}$$

The total heat flux is $q_{flux} = (T_{aw,eff} - T_{w,bulk})/R_{tot}$. Interface temperatures are derived sequentially via Fourier's law. Because σ depends on T_{hg}, the node iterates until the temperature residuals fall below a strict tolerance of 0.5 K. Finally, the bulk coolant temperature is stepped forward via a finite-difference enthalpy balance:

$$T_{w,i+1} = T_{w,i} + \frac{q_i \cdot 2\pi r_{avg} \cdot dx}{\dot{m}_w C_{p,i}} \tag{3.26}$$

3.3.5 Verification and Safety Criteria

To strictly guarantee steady-state operation and structural survival, the solver evaluates four specific criteria at every axial node.

1. Cavitation Avoidance:

Objective: Prevent the formation of vapor bubbles due to localized pressure drops, which would disrupt mass flow and cause severe pitting erosion on the internal manifold.

Criterion: The local static pressure must remain well above the saturation pressure of the bulk fluid, evaluated via Wagner-Pruss:

$$p_{static}(x) > SF_{cav} \cdot p_{vap}(T_{w,bulk}(x)) \quad \text{with } SF_{cav} \geq 1.4 \tag{3.27}$$

2. Onset of Nucleate Boiling (ONB) Two-Step Check:
Objective: Maintain predictable, single-phase forced convection. If boiling initiates, the relationship between wall temperature and heat flux becomes highly non-linear, invalidating the steady-state heat transfer measurements required by the assignment.
Criterion (Step 1 - Regime Check): The solver first evaluates the cold-side wall temperature (T_{wc}) against the local static saturation temperature (T_{sat}). If $T_{wc} < T_{sat}$, the wall is entirely subcooled, identifying a strictly safe single-phase forced convection regime.
Criterion (Step 2 - Superheat Check): If $T_{wc} \geq T_{sat}$, a localized superheated boundary layer exists. In this case, the Bergles-Rohsenow (1964) model [33] is evaluated to determine the required wall superheat necessary to activate nucleation cavities. The design is deemed safe if and only if the actual wall superheat ($\Delta T_{sup} = T_{wc} - T_{sat}$) is strictly lower than the activation threshold ($\Delta T_{ONB,req}$):

$$\Delta T_{ONB,req} = \sqrt{\frac{8\sigma_{LG} T_{sat} q_{flux}}{k_{water} h_{fg} \rho_V}} \tag{3.28}$$

3. Mechanical Stress Integrity:
Objective: Ensure the inner CuCrZr liner does not yield under the combined action of pressure and thermal gradients.
Criterion: In a regeneratively cooled jacket, the global chamber pressure is supported by the outer backing shell [22]. The inner liner is loaded exclusively by the differential pressure, $\Delta p = |p_{static,coolant} - p_{static,gas}|$. The maximum primary mechanical stress (σ_p) is the greater between the membrane hoop stress and the plate-bending stress at the fin roots:

$$\sigma_{mem} = \Delta p \frac{r(x)}{t_{wall}}, \quad \sigma_{bend} = \frac{1}{2}\Delta p \left(\frac{w_{ch}(x)}{t_{wall}}\right)^2 \tag{3.29}$$

This is superimposed with the biaxial thermal stress ($\sigma_{th} = \frac{\xi E \Delta T_{wall}}{1-\nu}$) using the Von Mises equivalent criterion:

$$\sigma_{VM} = \sqrt{\sigma_p^2 + \sigma_{th}^2 - \sigma_p \sigma_{th}} < \frac{\sigma_y(T)}{SF_{mec}} \tag{3.30}$$

where $\sigma_y(T)$ is the temperature-degraded yield strength of CuCrZr, evaluated at the mean wall temperature.

3.3.6 Sizing strategy for Thermal protection

The sizing procedure was initialized by defining a set of preliminary inputs, largely derived from current literature on cooling practices. The thickness of the CuCrZr chamber wall was set to 1 mm, to withstand the mechanical hoop stress for the between combustion gasses and the assumed 50 bar coolant pressure inside the jacket. The validation of structural margin is demonstrated in the final results section. An initial coolant mass flow rate ($\dot{m}_{H_2O}$) of 5 kg/s was assumed according to standard liquid rocket engine design guidelines for 100 kN thrust-class engines [3], the definitive mass flow rate is subsequently determined by the optimization algorithm. Initially, a steady-state thermal analysis was conducted to evaluate the temperature profile across the copper substrate and the fluid. The primary objective of this step was to evaluate the thermal conditions of the nozzle without any protective layer, to dictate wether to use a Thermal Barrier Coating (TBC). Once the solid stratigraphy was defined, these parameters served as inputs for the sizing of the channels.

3.3.7 Thermal Barrier Coating Analytical Sizing

A preliminary thermal analysis of the bare substrate wall was conducted. First, the hot-gas-side heat transfer coefficient (h_g) and the corresponding heat flux (q) were estimated using the Bartz equation [3], by imposing a structural limit temperature for the copper substrate of $T_{lim,wall} = 800$ K [11].
As shown in figure 3.1, this preliminary analysis yielded an extreme heat flux of ≈ 38 MW/m^2, which would lead the substrate to reach temperatures beyond its structural and operational limits. This would also push the feed systems to their operational limits and operate in nucleate boiling regime leading to a failure of the cooling jacket.

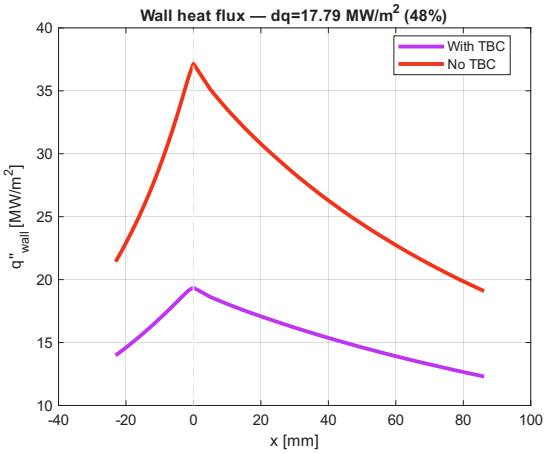


Figure 3.1: Heat flux with and without TBC

Consequently, the adoption of a Thermal Barrier Coating (TBC) is mandatory. The selected architecture is a Yttria-Stabilized Zirconia (YSZ) top coat, favored for its exceptionally low thermal conductivity and high-temperature stability [10]. To ensure proper adhesion a metallic bond coat thickness of 35 μm was assumed from table 1. In [12].
To initiate the preliminary sizing phase and guarantee safe single-phase operation, the target convective coefficient ($h_{c,\text{fix}}$) was deliberately constrained to a highly conservative baseline of $40,000$ W/(m^{2}K). It must be emphasized that this value does not represent an absolute thermodynamic limit for the cooling system, but rather a design choice to balance fluid-dynamic feasibility with manufacturing constraints [3]. By anchoring the sizing to this conservative $h_{c,\text{fix}}$, the mitigated heat flux (q_{new}) and the actual YSZ surface temperature (T_{surface}) were calculated as follows:

$$q_{new} = \frac{h_{c,\text{fix}}(T_{\text{lim,wall}} - T_{\text{water}})}{1 + h_{c,\text{fix}}\left(\frac{t_{\text{wall}}}{k_{\text{wall}}}\right)} \qquad (3.31) \qquad T_{\text{surface}} = T_{aw} - \frac{q_{new}}{h_g} \qquad (3.32)$$

Finally, the required thickness for the ceramic layer was computed by accounting for the thermal resistances of the imposed layers:

$$t_{TBC} = k_{TBC} \cdot \left(\frac{T_{surface} - T_{lim,wall}}{q_{new}} - \frac{t_{bond}}{k_{bond}}\right) \qquad (3.33)$$

This procedure was bounded by the metallurgical limit of the YSZ ceramic layer, which was conservatively set to 1673 K (1400°C) [10] to prevent phase degradation and melting.

3.3.8 Water Channel Sizing: Fluid-Thermal Model

First the solver predicts a surface temperature which then converges within a strict tolerance of 0.1 K. Upon convergence, the required local heat flux (q_{req}) in W/m^2 at each axial node is defined independently of the coolant mass flow rate.
By integrating the local heat flux over local areas, both the total thermal power absorbed by the entire jacket (Q_{tot}) and the throat section (Q_{to_throat}) were determined.

Now it is possible to compute the required convective heat transfer coefficient ($h_{c,req}$) that the coolant must sustain. This $h_{c,req}$ acts as a design target: it is the minimum convective efficiency required to dissipate the thermal load while keeping the underlying metallic layers and the TBC strictly within their operational temperature limits.
To model the actual heat extraction capabilities of the cooling jacket, the geometry was parametrized assuming N_{ch} rectangular milled channels. Each channel is defined by a variable width a_{ch} and height b_{ch}, and a fixed fin thickness of 1.5 mm. For a given geometry, the local fluid velocity $v_{water}(x)$ is computed through the continuity equation based on the local cross-sectional area.
The convective heat transfer coefficient provided by the water flow ($h_{c,real}$) is subsequently evaluated using the Dittus-Boelter correlation for turbulent flow in heated channels. While this correlation carries the known risk of overestimating the cooling capabilities, it is deliberately employed in this iterative optimization phase to avoid excessive complexity. The Reynolds (Re) and Prandtl (Pr) numbers are calculated evaluating water at the local bulk temperature, allowing the computation of the Nusselt number (Nu):

$$Nu = 0.023\, Re^{0.8} Pr^{0.4} \qquad (3.34) \qquad\qquad h_{c,\text{real}} = \frac{Nu \cdot k_{\text{water}}}{D_h} \qquad (3.35)$$

where D_h is the local hydraulic diameter of the channel and k_{water} is the thermal conductivity of the coolant. A specific geometric configuration is considered thermally valid only if $h_{c,real} \geq h_{c,req}$ along the entire nozzle axis[1].

3.3.9 Computational Optimization Algorithm

Since multiple geometrical configurations can satisfy the fundamental thermal requirement ($h_{c,\text{real}} \geq h_{c,\text{req}}$), the optimization consists in identifying the geometry that balances sustainable liquid velocities and manufacturing constraints. To address this, a multi-variable optimization algorithm identify the most efficient cooling channel topology.
The optimization routine is structured around three core logic blocks:

1. Design Search Space:
The algorithm performs a comprehensive sweep across a predefined geometrical and fluid-dynamic domain, exploring a 3D matrix of configurations by varying three independent variables: the coolant mass flow rate ($\dot{m}$), the number of channels (N_{ch}), and the channel height (b_{ch}). For every configuration generated in the search space, geometries that violate the following constraints are discarded:

- *Thermal Constraints:* The design is rejected if the computed cold-side temperature of the wall exceeds a margin of 15 K with respect to the water saturation temperature at a given pressure.
- *Manufacturing Constraints:* The design is rejected if the local channel width (a) falls below the minimum tolerances achievable via standard CNC milling processes.

2. Parametric Sensitivity and Dimensional Fix:
During preliminary optimization runs, a thermofluidodynamic behavior emerged regarding the channel height (b_{ch}). The algorithm consistently converged towards the absolute minimum allowable b_{ch} boundary (e.g., 2 mm). This is physically consistent with the

[1]The methodology and the formulas used for this computations where mainly derived from chapter 4.4 "Thrust chamber cooling" from Modern Engineering for Design of Liquid-Propellant Rocket Engines, Dieter K. Huzel and David H. Huang.

Dittus-Boelter model: reducing the channel height decreases the hydraulic diameter (D_h) and increases local flow velocity. Since the convective coefficient scales inversely with the hydraulic diameter ($h_c \propto D_h^{-0.2}$), shallower channels drastically enhance thermal efficiency.

3. Objective Function and Design Selection:

Among all the feasible configurations that survive the constraint filtering, the algorithm must select a single optimal design. The primary objective function was set to *minimize the total* ΔP. This optimization is crucial to reduce the required feed pressure and the related mass penalty of the propellant pumping system.
The mapping of the design space is visually represented in Figure 3.2. The gradient illustrates the total pressure drop, highlighting how lower velocities naturally correlate with lower pressure drops. The red line defines the *Feasibility Boundary*, the black line the manufacturing limit: All configurations below the red line and right to the black threshold are rejected by the algorithm due to violation of the set constraints.

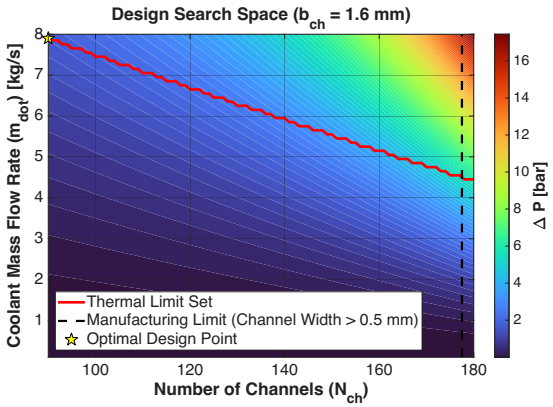


Figure 3.2: Design Search Space

The solver selected this specific point as it represents the minimum ΔP configuration capable of sustaining the required thermal margin not only at the throat, but across the entire diverging and converging sections.

3.4 External Supporting Apparatus and Pressurization System

The cooling-jacket circuit is not a flight subsystem, but a dedicated ground-test support system designed to provide a controlled and autonomous coolant supply during the whole firing sequence. According to the assignment, it must operate for the burn duration plus 2 s before ignition and 2 s after shutdown, dispose of the coolant after use, and remain independent from external utilities during operation. For this reason, the selected solution is a regulated pressure-fed water loop using compressed nitrogen as pressurant and a vented dump tank on the outlet side.
A regulated pressure-fed layout is appropriate because the key requirement is not thrust generation, but a reliable pressure cascade from storage to jacket inlet, including all line losses. A simple blowdown system would be mechanically simpler, but its pressure decay during the firing window would be undesirable for quasi-steady heat-load measurements.

3.4.1 Chosen architecture

The support apparatus relies on a sequential pressure cascade to drive the fluid. Specifically, the pressurization gas flows from a high-pressure nitrogen bottle through a pressure regulator, which maintains a constant driving pressure inside the downstream water tank. From this tank, the pressurized water is routed through a feedline equipped with a valve,

a cartridge filter, a calibration orifice, and the required piping elbows. The coolant then enters the engine cooling jacket. After absorbing the heat load from the thrust chamber, the water exits the jacket and passes through a downstream check valve before being finally discharged into a vented dump tank.
The coolant is therefore stored before the test, circulated once through the jacket, and disposed of after use. This closed-source / open-discharge configuration is simple, robust, and well suited to a short-duration experimental campaign.

3.4.2 Operating pressure levels and pressure cascade

The model adopts the following design values for the support apparatus: $p_{\text{tank}} = 64.0\ bar$, $p_{\text{N}_2,\text{init}} = 300\ bar$, $\Delta p_{\text{reg}} = 5\ bar$, $p_{\text{dump}} = 1.5\ bar$.

The corresponding final nitrogen pressure compatible with regulation is therefore:

$$p_{\text{N}_2,\text{final}} = p_{\text{tank}} + \Delta p_{\text{reg}} = 69.0\,\text{bar}.$$

The jacket inlet pressure is obtained from the external pressure budget. The total feedline pressure drop ($\Delta p_{\text{feedline}} = \Delta p_{\text{conc}} + \Delta p_{\text{dist}}$) is evaluated using the Darcy–Weisbach equation for distributed terms, whereas concentrated losses are grouped through equivalent K_i coefficients. With one valve, one calibration orifice, four 90° elbows and one cartridge filter, the computed external pressure loss directly yields the available jacket inlet pressure:

$$\Delta p_{\text{feedline}} \approx 11.83\,\text{bar} \qquad \Longrightarrow \qquad p_{\text{inlet}} = p_{\text{tank}} - \Delta p_{\text{feedline}} \approx 52.17\,\text{bar}.$$

This result is important because the relevant design quantity is not the storage pressure itself, but the pressure effectively available at the cooling-jacket inlet after all line losses. Along the jacket, the static pressure decreases from about 51.7 bar at the inlet to about 48.9 bar at the outlet. As shown in Figure 4.10, the pressure remains well above both the vapour-pressure curve and its safety threshold, confirming that the selected pressurization level is sufficient to overcome line losses while preserving single-phase operation.

3.4.3 Cavitation and hydraulic robustness

Cavitation was checked at the most critical external restriction, namely the calibration orifice. The minimum estimated pressure is about 45.7 bar, which remains orders of magnitude above the saturation pressure of water at the inlet temperature. The resulting cavitation margin is therefore extremely large. The same conclusion holds inside the jacket: the static pressure remains far above the local vapour pressure, and no critical tendency toward cavitation or boiling onset is predicted. Hence, the selected support apparatus does not merely deliver the required coolant flow rate, but does so in a condition that preserves the validity of the single-phase cooling assumptions used in the thermal model.

3.4.4 Autonomous sizing of the water tank

The support apparatus must remain autonomous during the entire test. Considering the motor burn time and the requirement of a 2 s margin both before ignition and after shutdown, the total operating time is:

$$t_b = \frac{I_{\text{tot}}}{F} = 25\,\text{s} \qquad \Longrightarrow \qquad t_{\text{op}} = t_b + 4\,\text{s} = 29\,\text{s}.$$

Given the design coolant flow rate, the total required water mass can be straightforwardly computed:

$$\dot{m}_w = 7.9\,\mathrm{kg/s} \qquad \Longrightarrow \qquad m_w = \dot{m}_w t_{\mathrm{op}} \approx 229.1\,\mathrm{kg},$$

which corresponds to a water volume of $V_w \approx 229.5\,\mathrm{L}$.
This is a substantial quantity, yet still realistic for a dedicated ground-test rig. More importantly, it shows that the support apparatus is no longer a minor accessory, but a major part of the experimental bench. The thermal choice of 7.9 kg/s therefore has a direct and non-negligible impact on facility size and logistics.

3.4.5 Sizing of the nitrogen pressurant bottle

The nitrogen bottle is sized assuming adiabatic blowdown, which is appropriate for the short duration of the test and conservative from a storage standpoint. The required initial gas volume is computed as

$$V_{\mathrm{N_2}} = \frac{V_w}{\left(\dfrac{p_{\mathrm{N_2,init}}}{p_{\mathrm{N_2,final}}}\right)^{1/\gamma_{\mathrm{N_2}}} - 1}, \qquad \gamma_{\mathrm{N_2}} = 1.4.$$

This gives $V_{\mathrm{N_2}} \approx 123.6\,\mathrm{L}$. The corresponding nitrogen mass is

$$m_{\mathrm{N_2}} = \frac{p_{\mathrm{N_2,init}} V_{\mathrm{N_2}}}{R_{\mathrm{N_2}} T_{\mathrm{gas}}} \approx 42.6\,\mathrm{kg}.$$

These values are fully compatible with a ground experimental setup, although they clearly indicate a sizeable support system. In equivalent spherical terms, the water tank would have a diameter of about 76.0 cm, whereas the nitrogen vessel would correspond to about 62.0 cm. The code also compares the adiabatic and isothermal assumptions: the latter would reduce the nitrogen requirement to about 68.6 L and 23.6 kg, but the adiabatic case is retained for design.

3.4.6 Design implications and final assessment

The previous calculations show that the support apparatus is feasible and internally consistent. The selected regulated pressure-fed architecture guarantees an inlet pressure close to 52 bar after external losses, preserves a large cavitation margin, and supplies the full coolant inventory required for the complete firing sequence without relying on external utilities during operation.
At the same time, the analysis highlights the engineering cost of the thermal solution: the selected coolant flow implies a sizeable support system, with about 230 L of stored water and roughly 124 L of nitrogen in the conservative adiabatic case. Therefore, the pressure-support hardware must be regarded as a major element of the experimental bench rather than as a secondary accessory.
In conclusion, the proposed configuration is hydraulically safe, compliant with the assignment constraints, and suitable for autonomous operation over the full 29 s experimental window.

Chapter 4

Results

4.1 Motor Design and Ballistics

4.1.1 Nozzle Design Outputs

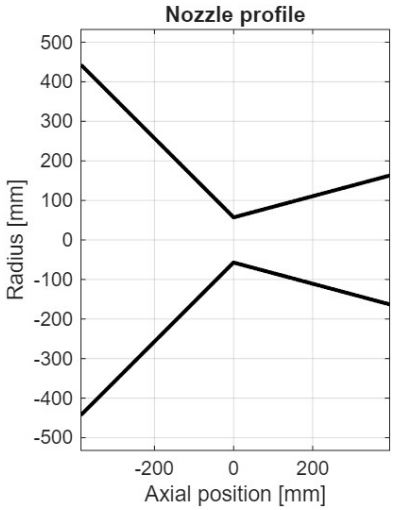


Figure 4.1: Geometric profile of the conical nozzle

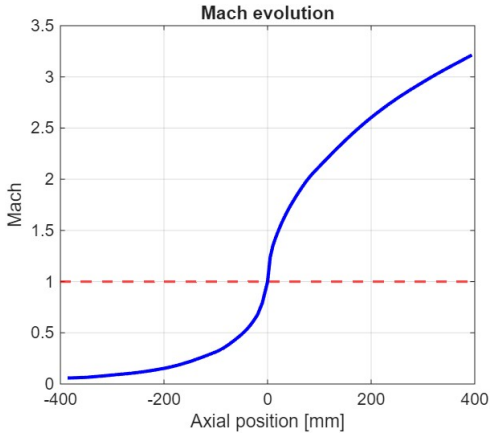


Figure 4.2: axial evolution of the Mach number

Figure 4.1 shows the nozzle profile, with the convergent section starting at the chamber diameter ($R_{\text{chamber}} = 443\,\text{mm}$, $\varepsilon_c \approx 60$) and converging to the throat radius ($R_t = 57\,\text{mm}$) over an axial distance of 386 mm. The divergent section expands from the throat to the exit radius ($R_e = 163\,\text{mm}$) over 395 mm. The convergent is steeper than the divergent due to the larger half-angle ($\beta = 45°$ vs $\alpha = 15°$). Figure 4.2 shows the Mach number evolution along the nozzle axis, obtained by interpolating the local flow velocity and speed of sound values provided by NASA CEA. The flow accelerates through the convergent, reaching $M = 1.0$ at the throat ($x = 0$). In the divergent section, the flow expands to $M \approx 3.2$ at the exit. The Mach profile in the convergent is relatively flat in the upstream region ($x < -200\,\text{mm}$), where the large cross-sectional area keeps the gas velocity low ($M < 0.2$). Initially, the viscous friction was computed assuming a truncated contraction ratio of 10. Once the BATES grain design was finalized, the skin friction integration was repeated using the actual chamber-to-throat area ratio ($\varepsilon_c \approx 60$). This extended integration domain resulted in a total friction force increase of less than 1%, definitively proving that viscous

friction effects in the deep, low-velocity convergent region are negligible. The resulting nozzle geometry is summarized in Table 4.1.

Table 4.1: Nozzle geometry and performance.

Parameter	Value	Unit
Throat area, A_t	102.6	cm^2
Exit area, A_e	833.1	cm^2
Initial mass flow rate, $\dot{m}$	42.42	kg/s

4.1.2 Final Grain Geometry

The converged grain dimensions are reported in Table 4.2.

Table 4.2: BATES grain geometry (single segment, $N = 1$).

Parameter	Value	Unit
Web thickness, w	202.2	mm
Outer radius, R	443.2	mm
Initial bore radius, r_i	241.0	mm
Segment length, L_0	1570.6	mm
$A_b(0) \approx A_b(w)$	32 475	cm^2

The grain geometry defined in Table 4.2 aligns with established solid rocket motor design principles, specifically optimized for static test configurations:

- **Length-to-Diameter Ratio (L/D):** The resulting ratio of ≈ 1.77 is characteristic of BATES (Ballistic Test Evaluation System) motors. While flight-optimized boosters typically exhibit higher L/D ratios (5–10) to minimize drag, a value between 1.5 and 2.0 is considered optimal for static tests to satisfy the neutrality constraint ($L_0 = 4r_i + 3w$) and ensure a stable pressure trace [3].
- **Volumetric Loading Fraction (VLF):** The grain design yields a Volumetric Loading Fraction (VLF) of approximately 70%. This value is lower than the typical range of 82%–94% suggested by Sutton [3] for flight-optimized motors. This discrepancy is justified by the specific mission requirements of the Stark-100k, which is a ground-test demonstrator rather than a mass-optimized flight unit [7, 18]. The priority was given to maintaining a quasi-neutral thrust profile through a BATES geometry, which inherently requires significant void volumes to allow for axial burning surfaces [9, 23]. Therefore, the trade-off between volumetric efficiency and combustion stability is consistent with the primary goal of nozzle heat-load characterization [11, 18].
- **Port-to-Throat Area Ratio (A_p/A_t):** The high initial ratio of 17.8 is a critical feature of the Stark-100k design. As erosive burning typically becomes significant only when A_p/A_t drops below 2–4 [4], this design maintains low internal gas velocities. This effectively eliminates erosive phenomena, ensuring that the burn rate remains strictly pressure-dependent and enhancing the overall predictability of the ballistic performance.

4.1.3 Time-Domain Results

Numerical integration over time generates the following results:

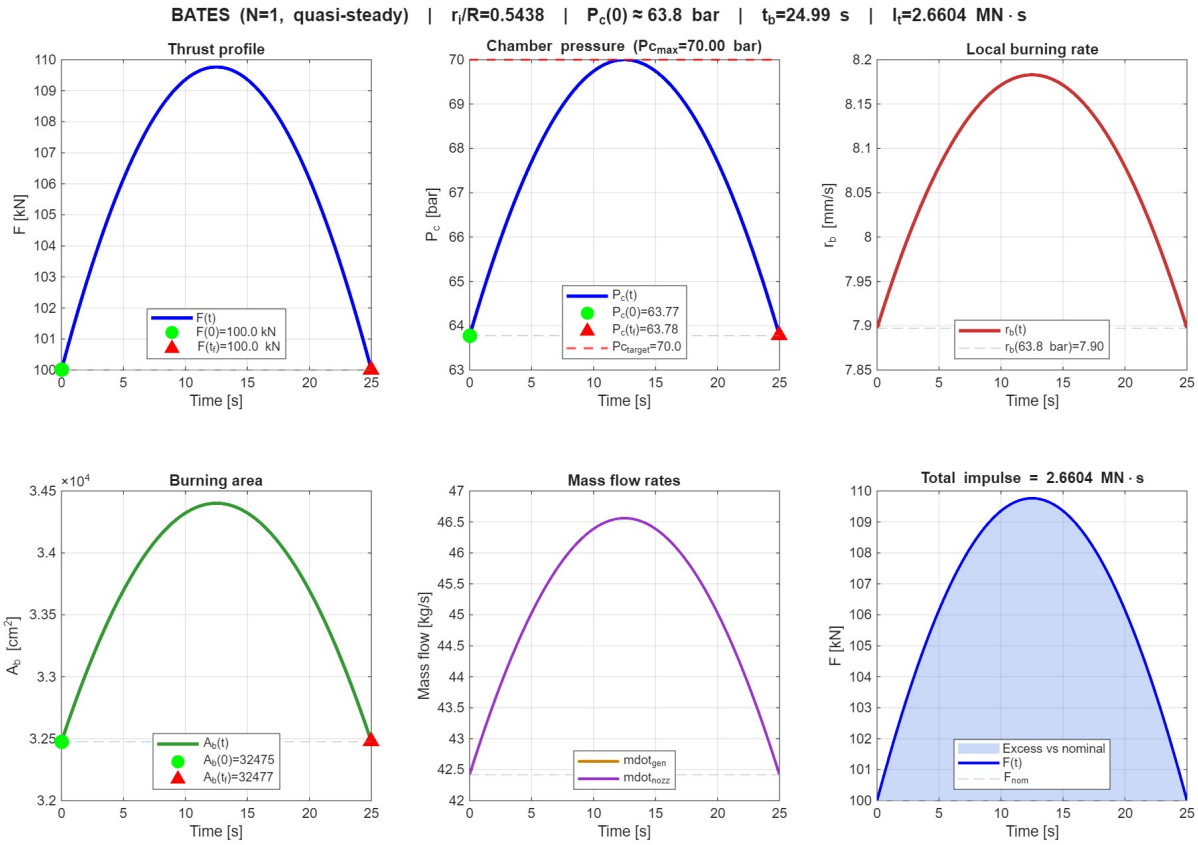


Figure 4.3: Time evolution of ballistic parameters

Figure 4.3 presents the time-domain profiles of the BATES motor. All quantities share the same symmetric hump shape because, in the quasi-steady model, they are all algebraic functions of the burning area $A_b(x)$: $P_c \propto A_b^{\frac{1}{1-n}}$, $r_b \propto P_c^n$, $\dot{m} = \rho_p r_b A_b$, $F = C_F P_c A_t$. Since $A_b(x)$ is a concave-down parabola peaking at mid-burn, the shape propagates to every variable. The BATES neutrality condition is confirmed by $P_c(0) \approx P_c(t_f)$ within 0.01 bar. The peak-to-initial pressure ratio is $70.0/63.8 \approx 10\%$, while the burning rate variation is only 3.6%, attenuated by the exponent $n = 0.38 < 1$. The quasi-steady simulation yields the performance summary in Table 4.3.
The total impulse, obtained by integrating the thrust profile over the burn duration, is:

$$I_{tot} = \int_0^{t_b} F(t)\, dt = 2.660 \text{ MN s} \tag{4.1}$$

This exceeds the 2.5 MN · s requirement by 6.4% because the BATES grain produces a quasi-neutral, not perfectly flat, thrust profile. This causes a mid-burn thrust hump ($F_{\max}/F_{nom} \approx 1.10$) that adds impulse above the nominal $F \times t_b$ baseline.

Table 4.3: Time-domain results from the quasi-steady BATES model.

Parameter	Value	Unit
Initial chamber pressure, $P_{c,in}$	63.77	bar
Maximum chamber pressure, $P_{c,max}$	70.00	bar
Initial thrust, $F(0)$	100.0	kN
Maximum thrust, F_{max}	109.8	kN
Burning rate (initial), r_b	7.90	mm/s
Burning rate (max), $r_{b,max}$	8.18	mm/s
Burn time, t_b	24.99	s
Propellant mass, M_p	1128.5	kg
Effective specific impulse, I_{sp}	240.4	s

4.1.4 Thermal Protection System Dimensioning

Following the methodology described in subsection 3.2.5, the thickness of the internal insulation was calculated to ensure the structural integrity of the motor throughout the combustion process. The design accounts for the high-temperature environment of the BATES motor and the erosive action of the exhaust gases. Table 4.4 summarizes the

Table 4.4: Combustion chamber insulation (EPDM + silica, $SF = 1.5$).

Zone	$\dot{r}$ [mm/s]	t_{insul} [mm]	Mass [kg]
Forward dome	0.08	3.0	2.2
Cylindrical wall	0.10	3.7	19.7
Aft dome (pre-nozzle)	0.15	5.6	4.2
Radial wall	–	1.5 (structural)	–
Total			**26.1**

required thicknesses and masses for the different sections of the combustion chamber, incorporating the safety factors previously discussed. Furthermore, the impact of the nozzle ablative insulation on the overall mass budget is detailed in Table 4.5.

Table 4.5: Nozzle ablative insulation (EPDM + silica, $SF = 1.5$).

Zone	$\dot{r}$ [mm/s]	t_{insul} [mm]	Mass [kg]
Convergent ($\varepsilon > 2$)	0.12	4.5	0.30
Divergent ($\varepsilon > 2$)	0.10	3.7	0.31
Nozzle total			**0.60**

The total thermal protection system mass is:

$$M_{TP} = M_{chamber} + M_{liner} + M_{nozzle} = 26.1 + 5.2 + 0.6 = 31.9 \text{ kg} \quad (2.8\% \text{ of } M_p) \quad (4.2)$$

The results show that the use of Silica-filled EPDM provides an effective thermal shield with a limited impact on the motor's mass fraction, maintaining the high performance required for the Stark Industries launch systems.

4.2 Cooling jacket

This section presents the analysis of the results obtained from the cooling jacket design. The main objective is to evaluate the thermal performance of the system, assessing its ability to provide adequate cooling of the nozzle walls under representative operating conditions. Temperature distributions, heat transfer effectiveness, and potential design criticalities are examined.

4.2.1 Heat loads

Results for heat flux and convective heat transfer coefficients are shown in Figure 4.4,Figure 4.5. The peak in both the convective heat transfer coefficient and the wall heat flux at the

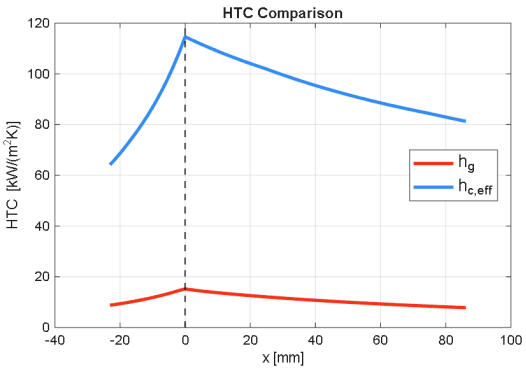


Figure 4.4: Comparison of convective heat transfer coefficients of hot gas and water

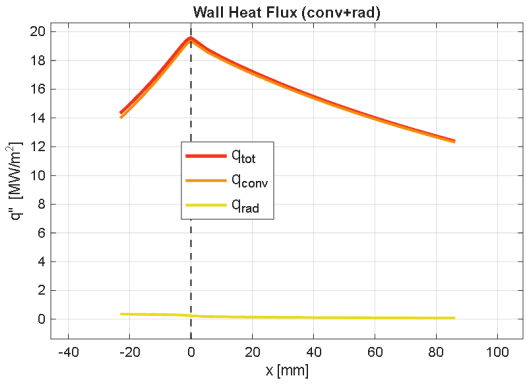


Figure 4.5: Wall heat flux axial evolution

throat is due to the extreme flow conditions in this region. Here, very high gas velocities and strong near-wall gradients enhance turbulence, increasing the heat transfer coefficient on the hot gas side. Even if the gas temperature is not at its maximum, the combination with a high coefficient produces a peak heat flux. Downstream, rapid supersonic expansion reduces density and temperature, lowering heat transfer, while upstream lower velocities and turbulence lead to reduced values. On the coolant side, the coefficient also peaks at the throat, where thermal load and temperature gradients are highest.

4.2.2 Material and layers thickness selection

The analytical sizing yielded an optimal YSZ thickness of 77 μm at the throat. As demonstrated by the verification code in Figure 4.6, this thickness represents an excellent design compromise: it successfully insulates the underlying metal, prevents the coolant from reaching boiling conditions, and maintains the ceramic surface safely below its critical temperature threshold. Finally, Figure 4.7 demonstrates how the presence of the TBC attenuates the gas-side heat transfer coefficient and subsequently limits the temperature rise of the coolant.

The table 4.6 summarizes the material selection and corresponding thicknesses adopted for both the structural components and the thermal protection of the nozzle, with which the previous heat transfer results have been obtained.

The primary insulation is provided by a layer of Yttria-Stabilized Zirconia (YSZ), which leverages a low thermal conductivity to establish a steep temperature gradient. Beneath

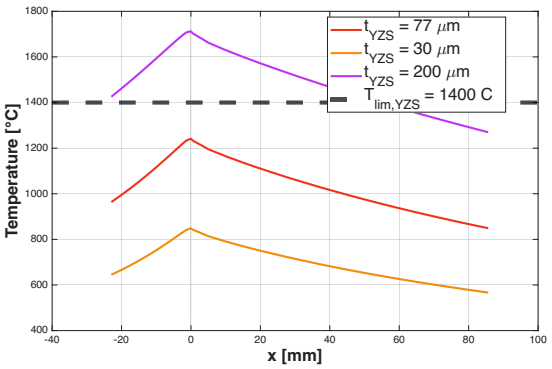


Figure 4.6: Effect of different YSZ thicknesses on its surface temperature

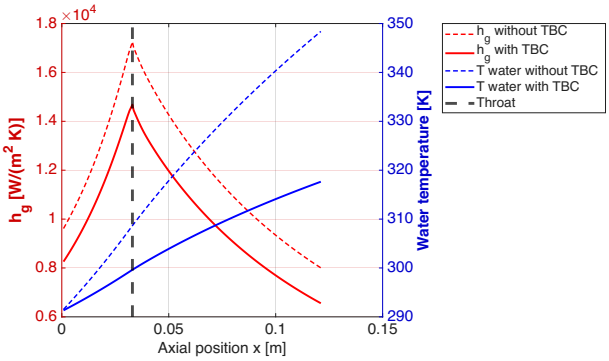


Figure 4.7: TBC effect on h_g and water temperature

	Top coat	Bond coat	Wall
Material	YSZ	NiCoCrAlY	CuCrZr
Conductivity $[\frac{W}{mK}]$	1.5	20	320
Thickness [mm]	0.077	0.035	1

Table 4.6: Materials specifications

this, the bond coat ensures chemical compatibility and adhesion between the ceramic topcoat and the substrate. Finally, a CuCrZr structural wall is employed; its high thermal conductivity facilitates efficient heat transfer to the coolant, ensuring that the wall temperature remains within the safe operating limits of the alloy. The thicknesses and thermal properties assigned to the coating layers correlate closely with empirical data: the TBC falls within the $< 100\ \mu m$ segment [20], while the bond falls within the range of $30 - 150\ \mu m$ [9], confirming the feasibility of the proposed architecture.

4.2.3 Channel geometry

The results from the optimization code for the final channels geometry are the one displayed in table 4.7.

Parameter	Symbol	Value
Mass flow rate	$\dot{m}_c$	7.9 kg/s
Channels	N_{ch}	90
Channel height	h_{ch}	1.6 mm
Width (throat)	w_{ch}^{t}	2.367 mm
Width (in/out)	$w_{ch}^{in/out}$	3.972 mm
Exit pressure	P_{out}	50 bar

Table 4.7: Cooling channel design parameters.

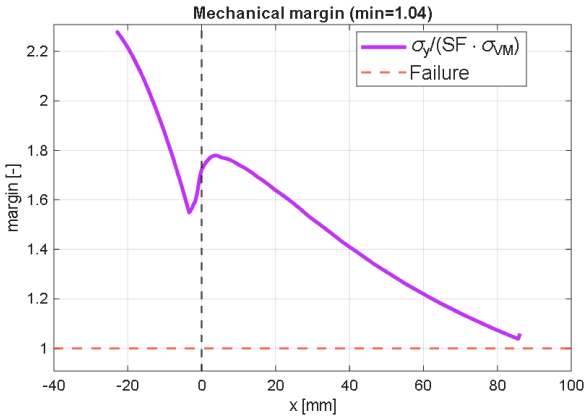


Figure 4.8: Mechanical margin evolution.

The operating pressure of the cooling system was selected to ensure single-phase conditions throughout the entire nozzle. In particular, it was chosen to remain consistently above the local vapor pressure of the coolant, thereby preventing cavitation. At the same

time, the pressure level was set such that the corresponding saturation temperature of water was always higher than the predicted wall temperature, in order to avoid the onset of boiling. Alongside thermal performance, mechanical integrity was a primary consideration. Structural robustness of the nozzle is illustrated in Figure 4.8, ensuring that the design withstands operational stresses without compromising the system.

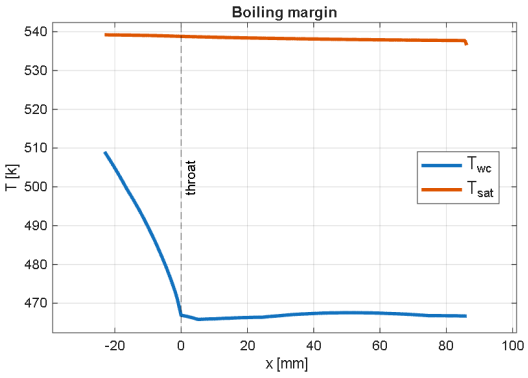


Figure 4.9: Comparison of the saturation temperature of the water in the working pressure condition with the temperature of the wall in contact with the water

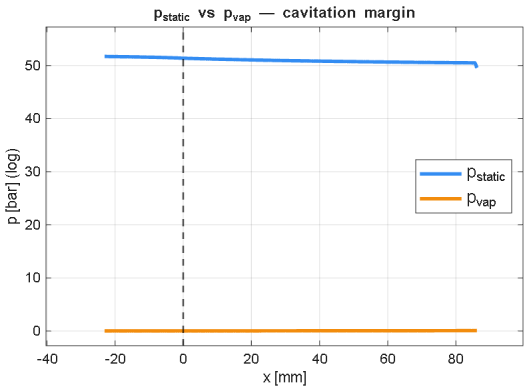


Figure 4.10: Operating pressure evolution along the channel compared with vapor pressure

4.2.4 Temperature profiles

The temperature profiles in Figure 4.11 show that all components remain within acceptable operating limits. In the throat region, where the heat flux reaches its maximum, the temperature of the TBC peaks as expected. However, thanks to its low thermal conductivity, the TBC effectively reduces the heat transmitted downstream, resulting in a significant temperature drop across its thickness. The bond layer follows the same spatial trend but at lower temperatures, as does the metallic wall, which experiences a further reduction in temperature, demonstrating that the combined TBC–bond system successfully shields the structural material. Finally, the cooling water temperature increases progressively along the flow direction due to heat absorption, but remains well below critical limits.

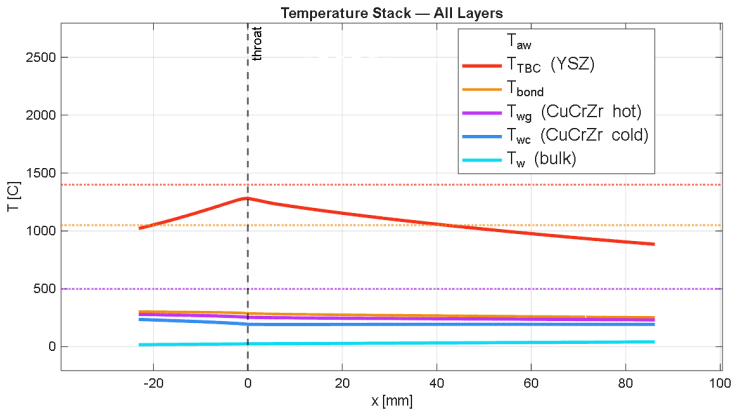


Figure 4.11: Temperature profile of every layer

Chapter 5

Considerations and Uncertainty Analysis

5.1 Uncertainty analysis

Our stochastic analysis is based on a Monte Carlo approach aimed at estimating the uncertainty affecting the maximum expected operating pressure (MEOP) and the burning time, under the assumption of constant ideal c^*. Assuming the grain inner and outer radii, the O/F ratio, and the initial chamber pressure $P_{c,\text{in}}$ as fixed, the parameters considered subject to statistical variability are the Saint Robert's law coefficients, the propellant density, the grain length, and the throat diameter. This choice was motivated by the fact that reasonable estimates of the statistical deviations of propellant density, grain length, and throat diameter can be derived directly from their nominal values and from the coefficients of variation reported in [8]. Accordingly, random values of propellant density, grain length, and throat diameter were generated from Gaussian distributions centred on their nominal values. This assumption is justified by the fact that these quantities are mainly affected by manufacturing tolerances and material variability, which can be interpreted as the cumulative effect of several small and approximately independent sources of uncertainty. Under such conditions, the Gaussian distribution represents a suitable first-order approximation of their statistical behaviour. On the other hand, the burn rate coefficient a and the pressure exponent n are treated as correlated random variables. It is appropriate to treat a and n as correlated variables because both are identified simultaneously from the same regression of the experimental burning-rate law. Therefore, the uncertainty affecting their estimates is not independent but intrinsically coupled through the fitting procedure. To account for this dependence, the covariance matrix associated with a and n was derived from the ballistic data provided. This made it possible to generate random pairs of Saint Robert's law parameters by sampling from a bivariate normal distribution centered on their nominal values. The Monte Carlo simulation employs these randomly generated values of propellant density and Saint Robert's law parameters to integrate the time evolution of chamber pressure and grain burning surface. This procedure is repeated 5000 times, each realization being associated with a different randomly generated set of a, n, ρ, L_0 and A_t. Repeating the simulation over a large number of realizations makes it possible to obtain the statistical distributions of the quantities of interest and, in the present case, to estimate the extreme values of maximum chamber pressure and burning time. The main results obtained from the Monte Carlo analysis can be summarised as follows:

1. maximum computed chamber pressure: $P_{MC,\max} \approx 79\,\text{bar}$;
2. maximum computed burning time: $t_{b,\max} \approx 27\,\text{s}$;
3. minimum computed burning time: $t_{b,\min} \approx 23\,\text{s}$.

These results indicate a possible increase in the maximum expected operating pressure, with

$$\frac{P_{MC,\max}}{P_{\max}} \approx 1.12,$$

corresponding to an increase of approximately 12% with respect to the nominal value of $P_{max} = 70\,\text{bar}$.
At the same time, the burning time is found to vary between approximately +8% and −8% relative to its nominal value of 25 s.

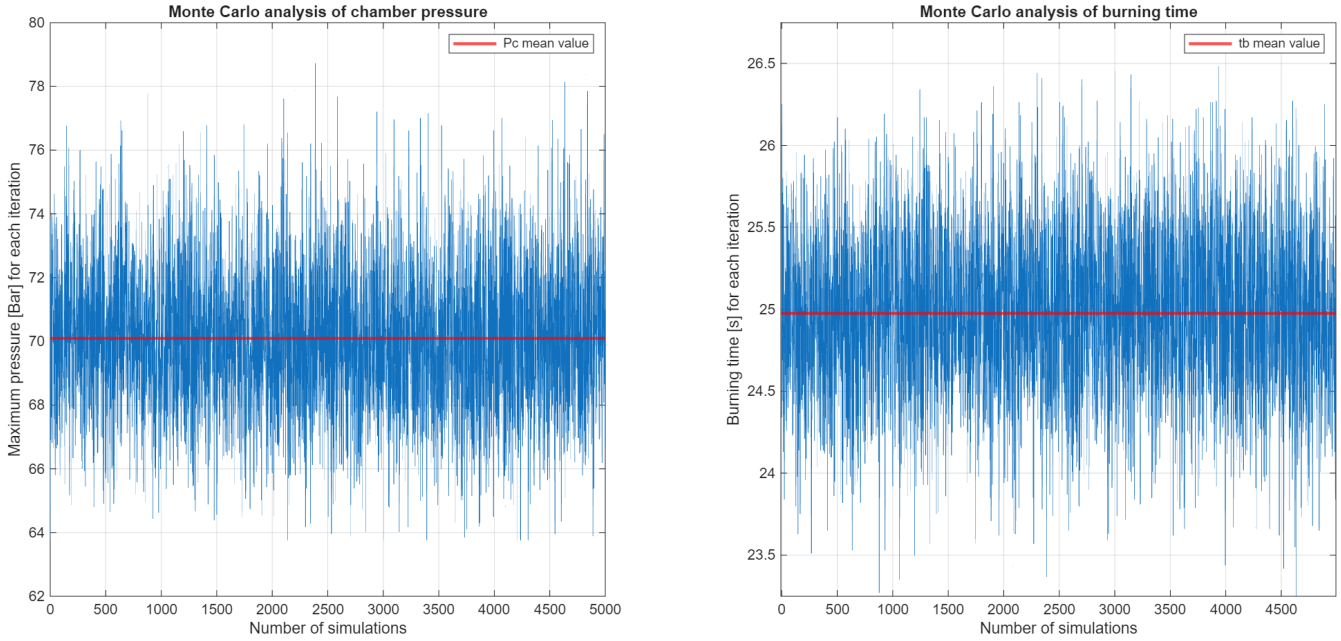


Figure 5.1: Monte Carlo results for chamber pressure and burning time.

The Monte Carlo analysis shows that the mean values of chamber pressure and burning time remain, as expected, very close to the corresponding nominal ones, while the associated dispersion is not negligible.
A more meaningful description of the stochastic response is provided by the relative standard deviations, which are found to be:

$$\sigma_{\text{rel},\,p_c} = \frac{100 * \sigma_{p_c}}{\mu_{p_c}} \approx 3.2\%, \qquad \sigma_{\text{rel},\,t_b} = \frac{100 * \sigma_{t_b}}{\mu_{t_b}} \approx 2\%.$$

The obtained values indicate a moderate but clearly appreciable propagation of uncertainty from the input parameters to the output quantities. The larger dispersion observed for chamber pressure is consistent with the structure of the adopted model, in which pressure depends nonlinearly on the ballistic and geometric parameters, while the burning time is influenced by the overall regression history of the grain and is therefore less sensitive to local variations. When compared with values reported in the literature, these relative standard deviations appear to be reasonably consistent. In particular, as shown in [19] and [18], the relative standard deviations of MEOP and t_b are generally of the order of a few percent.

5.2 Model considerations

The model adopted to estimate the main motor performance parameters is suitable for a preliminary design stage, as it enables rapid calculations without introducing excessive numerical complexity. Nevertheless, its main physical limitations should be acknowledged. First, the assumption of uniform chamber pressure neglects the possible presence of axial and local pressure gradients within the combustion chamber. Likewise, assuming a uniform burning rate over the entire propellant surface ignores the fact that the local regression rate may also depend on near-wall gas velocity, local temperature, heat flux, and other erosive effects, in addition to pressure. Another relevant simplification is the neglect of other possible sources of grain erosion and deviation from ideal behavior, such as geometric imperfections, non-uniform propellant composition, cracks, and density inhomogeneities. The frozen-at-throat approach is itself an approximation, since it assumes that the chemical composition of the combustion products remains unchanged downstream of the throat. In reality, recombination or dissociation processes may still occur in the divergent section, thus affecting the exhaust properties and nozzle performance. Finally, accounting only for divergence and friction losses as nozzle non-idealities may underestimate other performance penalties, such as flow separation, three-dimensional effects, and boundary-layer development. Since the engine serves as an experimental testbed rather than a flight-optimized system, the discrepancy does not affect the validity of the thermal data. The design achieves a total impulse of $2.66\,\mathrm{MN}\cdot\mathrm{s}$, exceeding the $2.5\,\mathrm{MN}\cdot\mathrm{s}$ requirement by 6.4%. This deviation is acceptable, as mission prioritizes heat load characterization at the nozzle throat. Since the engine serves as an experimental testbed rather than a flight-optimized system, the discrepancy does not affect the validity of the thermal data. Regarding the thermal management system, the sizing of the Thermal Barrier Coating (TBC) and the regenerative cooling jacket relies on standard empirical correlations, primarily the Bartz equation for the hot-gas side and the Dittus-Boelter equation for the coolant side. While these relations provide a reasonable approximation of the thermodynamic conditions experienced by the nozzle, they inherently carry significant margins of uncertainty. Historically, the Bartz correlation can exhibit deviations of approximately $\pm 20\%$ in predicting the hot-gas heat transfer coefficient, while the Dittus-Boelter equation carries an uncertainty of about $\pm 10 - 15\%$ [15] [21], which can further amplify under extreme heat fluxes and asymmetric heating conditions.

To balance computational cost and reliability, these simplified models were deliberately chosen for the iterative sizing strategy to avoid excessive complexity. Subsequently, more advanced and rigorous models were employed during the verification phase. This robust verification ensured that the geometric configuration derived from the simplified sizing yielded accurate thermal responses and physical consistency when compared to established literature.

Consequently, this conservative design philosophy, coupled with the inherent simplifications of the Dittus-Boelter correlation, inevitably leads to an oversizing of the cooling jacket. This overdesign manifests as high maximum mass flow rates and elevated coolant velocities within the channels, resulting in a heavier feed line that would be suboptimal for a flight-ready vehicle. However, given that the primary objective of this project is to develop a reusable test article rather than a flight-optimized engine, these penalties are considered acceptable. The design explicitly prioritizes the thermal tolerance and structural survivability of the metallic wall and the cooling jacket over system mass, ensuring safe and reliable operations during testing.

Chapter 6

Conclusions and future developments

6.1 Conclusions

The Stark-100k design successfully meets the primary objective of providing a stable platform for nozzle heat-load characterization. The BATES grain configuration ensured the required quasi-neutral pressure profile at 70 bar, despite a 6.4% deviation in total impulse (2.66 MNs) due to the characteristic thrust hump. This deviation, along with a 70% Volumetric Loading Fraction, is justified by the motor's experimental nature and the priority given to combustion stability over mass optimization. Thermal analyses confirmed that the water-cooling jacket, integrated with a YSZ coating, effectively replaces conventional refractory inserts while ensuring system reusability. Future work will focus on experimental static tests to correlate theoretical models with real-time temperature data.

6.1.1 Future Developments

To enhance physical fidelity and overcome current approximations while maintaining reasonable computational costs, future refinements should focus on the adoption of a quasi-one-dimensional chamber model to capture spatial variations in pressure and velocity, coupled with a burning-rate law corrected for local flow conditions. Furthermore, transitioning to thermochemical tools capable of handling actual HTPB properties, rather than the C_4H_6 surrogate, would provide a more realistic description of performance. The integration of non-ideal phenomena, such as throat erosion and boundary-layer development, is also recommended. Regarding the thermal management system, the use of rigorous correlations (e.g., Gnielinski) should be extended to the sizing phase and coupled with Computational Fluid Dynamics (CFD) to map localized thermal anomalies. For instance, a variable-topology channel architecture was explored during first analysis to optimize convergent section cooling. While 1D approximations showed theoretical benefits, severe fluid-dynamic uncertainties,such as unpredictable hot-spots and stagnation points, introduce systemic risks that currently outweigh the potential gains. Exploring this further will require much more complex models.

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